EA Prerequisite: Minimum Energy Performance

This prerequisite applies to

- BD+C: New Construction
- BD+C: Core & Shell
- BD+C: Schools
- BD+C: Retail
- BD+C: Data Centers
- BD+C: Warehouses & Distribution Centers
- BD+C: Hospitality
- BD+C: Healthcare

INTENT

To promote resilience and reduce the environmental and economic harms of excessive energy use and greenhouse gas emissions that disproportionately impact frontline communities by achieving a minimum level of energy efficiency for the building and its systems.

REQUIREMENTS

NC, CS, Schools, Retail, Warehouses & Distribution Centers, Hospitality, Healthcare

Comply with ANSI/ASHRAE/IESNA Standard 90.1–2016, with errata or a USGBC-approved equivalent standard.

ASHRAE 90.1-2016 Compliance pathways in Section 4.2.1.1 include compliance with all mandatory provisions, and compliance with one of the following:

- Prescriptive provisions of Sections 5 through 10
- Section 11 *Energy Cost Budget Method*
- Normative Appendix G *Performance Rating Method*. When using Appendix G, the Performance Cost Index (PCI) shall be less than or equal to the Performance Cost Index Target (PCI_t) in accordance with the methodology provided in Section 4.2.1.1. Document the PCI, PCI_t, and percentage improvement using metrics of cost or greenhouse gas (GHG) emissions.
- Exception to Mandatory Measures requirements: For ASHRAE 90.1-2016 mandatory provisions where the Appendix G *Performance Rating Method provides a methodology for demonstrating savings between the Proposed Building Performance (PBP) and the Baseline Building Performance (BBP)*, projects may model the *Proposed Building Performance* as designed in lieu of compliance with the mandatory provisions.
- Exceptional Calculations modeled in accordance with Section G2.5 may be modeled to document minimum prerequisite compliance.
- Only on-site or on-campus renewable energy that meets ASHRAE Standard 90.1-2016 Section G 2.4.1 requirements for on-site renewable energy may be used to meet ASHRAE Standard 90.1-2016 performance requirements.

GUIDANCE

The following guidance addresses both EA Prerequisite Minimum Energy Performance and EA Credit Optimize Energy Performance since the requirements and LEED documentation are closely linked for the Prerequisite and Credit.

Refer to the LEED v4 reference guide, EA Prerequisite Minimum Energy Performance for the referenced sections.

Beta Update

Updated referenced standards and added a new greenhouse gas emissions metric ensure that LEED continues to be a global leadership standard for energy performance and encourage owners to directly consider and address building carbon emissions.

Step-by-Step Guidance

ASHRAE Standard 209 provides a step-by-step methodology for applying energy modeling to inform the design process. Project teams are encouraged, though not required, to apply the guidance in ASHRAE Standard 209 as a best-practice approach for informing design through energy modeling. Following the guidance in Standard 209 will help project teams document achievement of LEED EA prerequisite Minimum Energy Performance Prerequisite, EA credit Optimize Energy Performance, and the energy modeling requirements for IP credit Integrative Process.

Step 1. Determine climate zone

Identify the project's climate zone according to ASHRAE 90.1–2016, Annex 1 (see *Further Explanation, Climate Zone Determination*).

Step 2. Review and address ASHRAE mandatory requirements

Early in the design process, review the mandatory provisions of ANSI/ASHRAE/IESNA Standard 90.1–2016, with errata (or a USGBC-approved equivalent standard for projects outside the U.S.). Read through Sections 5.4, 6.4, 7.4, 8.4, 9.4, and 10.4 to understand how the building design must respond to these requirements. Many mandatory requirements can easily be incorporated to the project in early design but are much harder to incorporate later in design and/or during construction.

- Typically, the architect is responsible for Section 5.4, Building Envelope; the mechanical engineer and plumbing designer are responsible for Sections 6.4, HVAC, and 7.4, Service Water Heating; and the electrical engineer is responsible for Sections 8.4, Power, and 9.4, Lighting. Compliance with Section 10.4 requires coordination across multiple disciplines.
- Ensure that the project complies with the mandatory measures throughout the design, construction, and commissioning process, particularly when major design decisions are implemented.
- Confirm that compliant components are included in the final construction documents.
- If compliance with ASHRAE 90.1-2016 mandatory provisions will be a hardship for the project, and the project intends to demonstrate compliance using the Appendix G Performance Rating Method, identify whether Appendix G Performance provides a method for quantifying savings of these mandatory provisions in the Proposed design as compared to the Baseline. For mandatory measures where Appendix G provides a methodology for demonstrating savings between the Proposed Building Performance (PBP) and the Baseline Building Performance (BBP), in lieu of compliance with the mandatory provisions, the Proposed Building efficiencies and controls may be modeled as designed, while the Baseline Building efficiencies and controls are modeled consistent with ASHRAE 90.1-2016 Appendix G requirements. Examples include:
    - In lieu of complying with mandatory daylighting and occupancy sensor control requirements, document the proposed model without credit for lighting controls, with lighting schedules modeled identically to the baseline model.
    - If the exterior lighting power allowance exceeds the mandatory allowance, model the proposed exterior lighting power as designed, and model the Baseline consistent with Appendix G.
    - If the HVAC efficiency does not meet the mandatory provisions, model the proposed efficiency as designed, and model the Baseline consistent with Appendix G.
- The following additional exceptions to mandatory measures may also be applied:
    - Section 5.4.3.4 Vestibules: The project may apply ASHRAE 90.1-2016 Addendum bf, Section 5.4.3.4, which adds exceptions for:

        - Self-closing doors in buildings in Climate Zone 0, 3, and 4 that have an air curtain complying with Addendum bf, Section 10.4.5.
        - Self-closing doors in buildings 15 stories or less in Climate Zones 5 through 8 that have an air curtain complying with Addendum bf, Section 10.4.5.

    Note that the air curtain energy must be included in the Proposed design model and not in the Baseline if applying the Appendix G Performance Rating Method.

    - Section 8.4.2 Automatic Receptacle Control:
        - Path 1. Projects Using the Appendix G Performance Rating Method may model a penalty in the energy model for the spaces where mandatory ASHRAE 90.1 receptacle controls are not implemented. The following modeling requirements apply:
            - The Proposed receptacle power density modeled for these spaces shall be the greater of 0.75 Watts per square foot (8.1 Watts per square meter) or the design coincident peak receptacle power density (if known)
            - The receptacle schedule modeled in the Proposed design for these spaces shall have a minimum Equivalent Full Load Hours of operation no less than: the ASHRAE 90.1-2016 User's Manual default schedule for office occupancy (2,920 Equivalent Full Load Hours per year); or 120% of the occupied hours of operation for the facility; or detailed justification shall be provided supporting an alternate schedule.
            - The Proposed model shall include either a 20% increase in the receptacle power density for these spaces OR a 20% increase in the scheduled receptacle Equivalent Full Load Hours of Operation versus the Baseline model.
        - Path 2: Projects must demonstrate that the project has implemented efficiency measures that will achieve an equal or greater reduction in receptacle energy consumption and will persist for a similar timeframe to those achieved by ASHRAE 90.1-2016 Section 8.4.2. It is recommended that a Credit Interpretation Request be submitted when pursuing this approach. The project must provide documentation regarding the receptacle equipment controls that will be implemented for the project; and must provide justification supporting the claim that the savings over the life of the efficiency measure will be similar to those anticipated for a project compliant with Section 90.1-2016 Section 8.4.2. Note: the project is not eligible for any further receptacle savings in these spaces using the Appendix G Exceptional Calculation Method or using the Prescriptive path for measures relying on ENERGY STAR eligible equipment features.

(See Further Explanation, Table 1. Changes in ASHRAE 90.1 mandatory requirements, 2010 to 2016).

Step 3. Identify energy use target for building

This step is required for all projects pursuing credit under IP credit Integrative Process and recommended for all other projects.

Set an energy goal for the project early in the design process. Identifying an energy goal can help prioritize efficiency strategies, integrate systems, reduce first costs, and improve building performance.

For IP Credit Integrative Process, the target must be established using one of the following metrics:

- kBtu per square foot-year (kWh per square meter-year) of site energy use
- kBtu per square foot-year (kWh per square meter-year) of source energy use
- pounds per square foot-year (Kg per square meter-year) of greenhouse gas emissions
- energy cost per square foot-year (cost per square meter-year)

For building types such as manufacturing, if a different metric is more appropriate for benchmarking building energy consumption (e.g. kBtu per pound of finished product (kWh per kilogram of finished project)), project teams may use that metric in lieu of the metrics above. When using a different metric, provide a brief narrative supporting that the metric used is a more appropriate means of benchmarking building energy consumption for the building type and function.

Consider using ENERGY STAR's Target Finder to develop the EUI goal that will meet the credit requirements.

Consider applying the guidance in ASHRAE Standard 209 Section 5.4 and Informative Appendix B (Benchmark Information) when establishing the energy goal for the project.

Step 4. Select option for credit compliance.

Select the appropriate option in EA credit Optimize Energy Performance for the project (see *Further Explanation, Selecting an Option*). Review the requirements for EA credit Optimize Energy Performance before making a selection.

- Option 1. Energy Performance Compliance is available to all projects. This option is the best method for informing design decisions throughout the design process, and has the greatest number of points available under EA credit Optimize Energy Performance. For projects using this method, a Baseline Building Performance Model and Proposed Building Performance model are developed consistent with ASHRAE 90.1-2016 Appendix G, Performance Rating Method.
- Options 2 and 3 are for projects intending to apply simple upgrades to mechanical, envelope, lighting, appliances, and/or process equipment. Projects with minimal scope, such as Core and Shell projects with scope limited to the building envelope and exterior lighting often apply this option, since prerequisite compliance may be documented based solely on the project scope, and Optimize Energy Performance credit may be documented for the specific elements that are within the project scope.

  Projects must demonstrate compliance with EA prerequisite Minimum Energy Performance using the ASHRAE 90.1-2016 prescriptive compliance pathway to apply these options. Projects pursuing this option should work with the architect and engineers to assess the prescriptive requirements of ANSI/ASHRAE/IESNA Standard 90.1–2016, with errata (or a USGBC-approved equivalent standard for projects outside the U.S.) and ensure that the design will comply with envelope, HVAC, service water-heating, and lighting requirements, per Sections 5.5, 6.5, 7.5, 9.2.2 for all elements within the project scope of work. Compliance with prescriptive ASHRAE 90.1-2016 requirements and prescriptive EA credit Optimize Energy Performance requirements should be verified early in the design process, with ongoing verification of compliance throughout the design and construction process.
  - Option 2: Project teams may pursue a limited number of points under EA credit Optimize Energy Performance. The eligible project types for Option 2 include the following:
    - Small to medium office buildings, less than 100,000 square feet (9 290 meters)
    - Medium to large box retail buildings, 20,000 to 100,000 square feet (1 860 to 9 290 square meters)
    - School buildings, any size
    - Large hospitals, more than 100,000 square feet (9 290 square meters)
    - Grocery stores
  - Option 3 is a prescriptive option available for projects with less than 2,000 square feet (186 square meters) of data center space, laboratory space, or manufacturing space.

- For projects using the BD+C Data Center rating system, Option 4 may be used to demonstrate system optimization of the Data Center Mechanical and Electrical equipment using ASHRAE 90.4 2016.
- If the project is not pursuing any points under EA credit Optimize Energy Performance, the project may demonstrate EA prerequisite Minimum Energy Performance compliance using ASHRAE 90.1-2016 Section 11 Energy Cost Budget. This option uses energy modeling with trade-offs but has different Baseline building modeling requirements than the normative Appendix G performance rating method.
- Refer to the LEED Credit Library *Pilot alternative compliance paths (ACP)* for additional options for demonstrating EA Prerequisite Minimum Energy Performance and EA Credit Optimize Energy Performance. Pilot ACPs test credits that achieve a similar intent to the referenced credit by applying a new strategy. For EAp Minimum Energy Performance and EAc Optimize Energy Performance, these may include alternative metrics for evaluating decarbonization and energy efficiency using ASHRAE 90.1-2016 Appendix G, entirely new compliance paths intended to streamline the analysis and documentation process, or compliance paths that reference alternative standards.

Step 5. Develop preliminary energy model or alternate energy analysis

To achieve EA credit Optimize Energy Performance, project teams must analyze efficiency measures during the design process, focusing on load reduction and HVAC-related strategies or passive measures appropriate for the facility, and account for the results during design decision making.

For projects using Option 1. Energy Performance Compliance, the best approach for analyzing efficiency measures is a preliminary energy model, which evaluates heating and cooling load reduction strategies, passive HVAC strategies, and HVAC efficiency and control strategies (see *Further Explanation, Developing a Preliminary Energy Model* and *Further Explanation, Modeling HVAC Systems)*. ASHRAE Standard 209 provides a standardized methodology which may be used for developing preliminary energy models that are used to inform the design process (See ASHRAE 209 Sections 6.3 - Load Reduction Modeling and 6.4 - HVAC System Selection Modeling).

- Developing an early model of the proposed design will help the design team explore the energy consequences of design options and will provide an early estimate of energy performance.
- When evaluating energy usage in different scenarios, consider strategies for lighting and daylighting, envelope, orientation, and passive conditioning and ventilating systems, in terms of projected energy savings and capital costs as they relate to all building systems. If pursuing the Integrative Process Credit, evaluate these parameters at a concept level early in design.

Project teams may also use past energy analyses of similar buildings or published energy modeling results, such as the ASHRAE Advanced Energy Design Guides (AEDGs) to guide decision making in lieu of a preliminary energy model, though the results will be less project-specific. The AEDGs were designed around specific building types and sizes by climate zone, making the recommendations most appropriate for projects with attributes similar to those specified types, sizes, and locations.

Step 6. Ongoing iterations of Design Phase Energy Model (Option 1. Energy Performance Compliance), or Prescriptive compliance documentation (Options 2 or 3).

Option 1. Energy Performance Compliance

For projects pursuing EA credit Optimize Energy Performance Option 1. Energy Performance Compliance:

Once the HVAC system and other design parameters are established, build or update the proposed building energy model to reflect the anticipated design (see *Further Explanation, Building the Proposed Energy Model*).

- Update the proposed model to reflect changes that occur throughout the design process to optimize energy performance and assist with design decisions.
- Ensure that all efficiency strategies are analyzed well before design documents are finalized.
- For elements or systems that cannot be readily modeled by the software or to document credit for unregulated loads, use the Exceptional Calculation Method (see *Further Explanation, Exceptional Calculation Method* and v4 Reference Guide, *Further Explanation, Common Issues with Energy Modeling*).
- Document credit for qualifying renewable energy in the proposed energy model (see *Further Explanation, Applying Renewable Energy Savings*)

For projects pursuing EA credit Optimize Energy Performance, Option 1. Energy Performance Compliance:

Build a baseline model that reflects the minimum requirements according to ASHRAE 90.1–2016, Appendix G (see *Further Explanation, Building the Baseline Performance Model*).

- When modifications are made to the proposed energy model, update the baseline accordingly.
- Consider constructing the baseline model early in the design process so that the design team can see the effect of design changes on the percentage savings relative to ASHRAE 90.1. This will contribute toward achieving more points under the related credit.
- Use the Minimum Energy Performance Calculator to help create the baseline model. This tool was designed to help project teams create a baseline model in alignment with Appendix G requirements.

Update the proposed energy model as necessary to reflect final construction details and specifications and make any necessary corresponding updates to the baseline model.

Use the results from the baseline and proposed models and the Building Performance Factor to determine the anticipated energy cost and greenhouse gas emissions savings (see *Further Explanation, Calculations*, *Energy Cost* and *Greenhouse Gas Emissions*). Either the cost or the GHG emissions metric may be used to show prerequisite compliance.

Prerequisite Compliance Only – ASHRAE 90.1-2016 Section 11 Energy Cost Budget

For projects that are not pursuing EA credit Optimize Energy Performance, and are documenting compliance using ASHRAE 90.1-2016 Section 11, Energy Cost Budget, complete the ASHRAE 90.1 Section 11 design energy cost and energy cost budget models, and complete the ASHRAE 90.1 ECB forms demonstrating compliance.

Option 2, 3, or 4. Prescriptive Compliance

Prepare final ASHRAE 90.1 documentation confirming compliance with the mandatory and prescriptive requirements of ASHRAE 90.1-2016.

For projects pursuing Optimize Energy Performance, see Step-by-Step Guidance, Prescriptive Compliance.

Further Explanation

Calculations

For projects using ASHRAE 90.1-2016 Appendix G, Performance Rating Method, the following equations apply:

Equation 1. Section G1.2.2: Performance Cost Index

Performance Cost Index = Proposed building performance / Baseline building performance.

where Proposed building performance and Baseline building performance are calculated in accordance with ASHRAE 90.1-2016 Appendix G.

Equation 2. Section 4.2.1. Performance Cost Index

PCI_t = [BBUEC + (BPF x BBREC)] / BBP

where:
PCI = Performance Cost Index calculated in accordance with ASHRAE 90.1-2016 Section G1.2 as described above.
BBUEC = Baseline Building Unregulated Energy Cost, the portion of the annual energy cost of a baseline building design that is due to unregulated energy use.
BBREC = Baseline Building Regulated Energy Cost, the portion of the annual energy cost of a baseline building design that is due to regulated energy use.
BPF = Building Performance Factor from Table 4.2.1.1. For building area types not listed in Table 4.2.1.1 use "All others." Where a building has multiple building area types, the required BPF shall be equal to the area-weighted average of the building area types. The Building Performance Factors from Table 4.2.1.1 (BPF) represent the average ratio of ASHRAE 90.1-2004 versus ASHRAE 90.1-2016 regulated energy cost for a given building type and climate. For example, a BPF of 0.59 represents an ASHRAE 90.1-2016 regulated energy cost that is 59% of the 90.1-2004 regulated cost for the given building type and climate. Table 4.2.1.1 lists
BBP = Baseline Building Performance.

Equations 1 and 2 adjustments for greenhouse gas emission metric:
When using Greenhouse gas emissions as the metric, replace all ASHRAE 90.1-2016 references to cost with Greenhouse Gas Emissions (CO2 equivalent emissions). Rather than using utility rates, use the appropriate greenhouse gas emission coefficients for each energy source (See *Further Explanation, Greenhouse Gas Emissions*)

Total LEED points for EAc Optimize Energy Performance are determined by summing the points documented in Table 1 (cost metric) PLUS the points documented in Table 2 (GHG emissions metric), where points for each Table are determined based on the project percent improvement PCI below the PCI_t documented for each metric.

For Table 1:

- Percent cost PCI below PCI_t = $1-PCI/PCI_t$

For Table 2, calculate PCI and PCI_t using greenhouse gas emissions instead of cost.

- Percent GHG Emissions PCI below PCI_t = $1-PCI/PCI_t$

If a project consists of a combination of New Construction and either Major Renovation or Healthcare, use Equation 1 to determine the appropriate percentage improvement target from Table 1 or Table 2 (points for percentage improvement in energy performance).

Equation 3. Target energy savings for combination of New Construction, Major Renovation, and/or Healthcare (HC).

Target percentage = {(existing or HC floor area / total floor area) x target percentage of savings for Major Renovation or Healthcare} + {(new non-HC floor area / total floor area) x target percentage of savings for New Construction}

Climate Zone Determination

Determining the right climate zone for the project is essential since the requirements are specific to each climate zone. ASHRAE 90.1–2016 defines eight climate zones (Miami is in climate zone 1; Anchorage is in climate zone 8) and three climate types: A (moist), B (dry), and C (marine).

To find the project's climate zone and type, consult ASHRAE 90.1–2016, Annex 1. For projects in the U.S, refer to the appropriate state and county in Table Annex 1-1. For projects in Canada, refer to the province and location in Table Annex 1-2. For locations outside of the U.S. and Canada, refer to the closest or most similar location in Table Annex 1-3. International projects may also refer to ASHRAE Standard 169-2020 to determine the project's climate zone based on historical weather data for the project's location.

Selecting an Option

Determining which option is most appropriate for the project requires knowing the extent of energy performance feedback desired during the design process.

- If detailed feedback is important during the design process, or the project is targeting a high level of energy performance and low greenhouse gas emissions, then the performance option (1) is most appropriate. Energy modeling generates information on the potential savings associated with various efficiency measures, both in isolation and in combination with other measures. Often this includes estimates of overall energy use, greenhouse gas emissions, or cost savings for the project, which can help gauge progress toward an energy savings and greenhouse gas emissions goal or achievement of points under the related credit.
- If the owner or design team requires only limited feedback, and the project is not targeting significant energy savings then one of the prescriptive options (2, 3, or for Data Centers - 4) may be more appropriate. These options are best suited for projects with standard systems and provide only limited feedback, in that all efficiency measures must be incorporated to achieve the prescribed threshold for energy performance.

Performance Path

The following factors could indicate that Option 1 would be advantageous to the project:

- The project is targeting a high level of energy performance
- None of the Optimize Energy Performance prescriptive pathways are available to the project because of the building's type or size.
- The project has an HVAC system that is not covered by one of the prescriptive options.
- The project team wants to explore the energy performance and load reduction effects of several envelope and lighting designs and mechanical systems.
- The project team is planning to maximize the number of points available through EA credit Optimize Energy Performance.
- The project team wants to achieve efficiency trade-offs between systems, offsetting the lower efficiency of one system by the improved efficiency of another.
- The owner is interested in commercial building federal tax credits or state, local, or utility incentives that require energy modeling. The modeling requirements for such incentive programs may be different from the ASHRAE 90.1–2016 requirements, however.
- The owner wants an estimate of the carbon reductions or lower operating costs (energy savings, demand charge savings) from energy strategies, beyond a simple calculation for individual energy conservation measures.

Project teams pursuing Option 1 should consider referencing ASHRAE Standard 209-2018, Energy Simulation Aided Design for Buildings except Low Rise Residential Buildings, which defines best practices and minimum requirements for providing energy design assistance using building energy simulation and analysis.

Before undertaking energy modeling as part of the performance path, consider the timing of the simulation preparation and presentation, and understand the costs and benefits of energy modeling as it relates to the project. When energy modeling is conducted late in design, its value is very limited, except as a compliance tool: the model can only estimate the energy savings of the design.

In contrast, if initiated early and updated throughout the design process, energy modeling can be a decision-making tool, giving feedback as part of the larger analysis of building systems and components. The best value will be seen when energy modeling is used as a tool in an integrated design process because it enables a more informed, cost-effective selection of efficiency strategies.

Note: Early design phase analysis is required to earn points under EA credit Optimize Energy Performance.

Develop clear expectations for the presentations of modeling results and their integration into the project schedule. Ideally, iterations of the model will be presented to the team during each stage of design, beginning as early as possible, when the project goals are incorporated into preliminary plans. Updates should be presented as the design is developed further to incorporate engineering and architectural details, and again when the construction documents are being prepared.

Regardless of the project design phases, energy modeling can still be performed as the design progresses. However, the potential benefit of energy modeling decreases as the design becomes finalized and opportunities for incorporating changes are lost. Ask the project's energy modeler to provide a schedule that integrates energy modeling into the design process, with appropriate milestones.

To develop an accurate and compliant energy model, it is important that the energy modeler read and understand ASHRAE 90.1–2016 (Appendix G and all mandatory measures from each Section in particular) in its entirety, not just the portions that apply to the project. This will enable a more complete understanding of the energy modeling protocols and methodologies required for LEED projects (see *Further Explanation, ASHRAE 90.1, 2016 versus 2010*). The energy modeler should also consider reading the ASHRAE 90.1–2016 User's Manual, which provides examples and further guidance relevant to Appendix G.

Prescriptive paths
The following factors could indicate that Option 2 or Option 3 would be advantageous to the project:

- The project budget and timeline would benefit from simplified decision making and analysis during the project design.
- The additional cost of energy modeling would not be warranted.

Although the prescriptive paths are applicable to some large or complex projects, such as hospitals, they were designed primarily for smaller projects, for which the cost of energy modeling would represent a high percentage of the project budget.

The prescriptive paths are available only for projects that meet certain criteria. Review the project's eligibility for the ASHRAE 50% Advanced Energy Design Guides and/or Option 3. Systems Optimization, and Option 4: Data Centers Only – System Optimization. If these prescriptive option do not fit the project type, the team must pursue Option 1 in order to achieve points under EA credit Optimize Energy Performance.

If the project is eligible for both of the prescriptive options, determine which is more appropriate based on the specific option requirements as well as future credit goals. The building type, for example, may not match those in the AEDGs, or the Option 3 prescriptive requirements may align better with the project's goals and design.

Option 2. ASHRAE 50% AEDG, delivers a 50% savings over ASHRAE 90.1–2004 when all requirements in all categories are met. Have the mechanical engineer review the applicable AEDG requirements for the project type. If the project is expected to have unique systems, potential equipment is not listed, or the system capacity is not likely to fall within the ranges in the AEDG, then the project team cannot pursue Option 2, and must pursue Option 1, Option 3, or (for Data Centers) Option 4.

Option 3. Systems Optimization savings vary dependent on the efficiency strategies, climate zone and building type.

Energy Modeler Qualifications

Refer to the LEED v4 reference guide.

Developing a Preliminary Energy Model

Refer to the LEED v4 reference guide, with the following addition:

- See ASHRAE 209 Sections 6.3 - Load Reduction Modeling for further guidance.

Modeling HVAC Systems

Refer to the LEED v4 reference guide, with the following addition:

- See ASHRAE 209 Section 6.4 - HVAC System Selection Modeling for further guidance.

Building the Proposed Building Performance Model

A team that has already prepared a preliminary model may update it to reflect the newest design information throughout the project.

Create or update proposed building characteristics based on the latest information and specifications on systems, assemblies, and equipment in the current design. This can be accomplished as early as design development to estimate projected savings, and later updated when the construction documents are complete. Then analyze remaining efficiency strategies that the team would like to consider before the design documents are finalized. For example, the proposed building performance energy model could be used to evaluate the performance and cost implications of value engineering decisions.

In most cases, the ASHRAE 90.1 proposed building performance model will exactly mirror the project design. However, ASHRAE 90.1 Appendix G indicates some specific cases where the modeled parameters may vary from the actual design. Examples include:

1. All conditioned spaces in the proposed design, with the exception of a few space types must be simulated as being both heated and cooled even if a heating or cooling system is not installed (Table G3.1(Proposed)(1)(b)).
2. HVAC fans used for ventilation shall be cycled on and off to meet heating loads during unoccupied hours, even if the systems are scheduled to remain off during unoccupied hours in the project design (Table G3.4(Proposed)(4)).
3. Lighting in unfinished spaces shall be modeled as meeting ASHRAE 90.1-2016 Table 9.5.1 prescriptive requirements.

Building the Baseline Building Performance Model

Developing the baseline building performance model is a detailed process that requires a good working knowledge of ASHRAE 90.1–2016, Appendix G. The baseline model represents a typical design for a building of the same size, function, and number of floors as the proposed building. It meets the prescriptive and mandatory requirements of ASHRAE 90.1-2004 for a building with standard practice HVAC, lighting, plumbing and envelope systems.

In general, baseline building performance energy model development begins by changing the inputs for all the components, assemblies, systems, and controls of the proposed building performance energy model to values, types, and controls prescribed in accordance with 90.1–2016 Appendix G. Whereas previous versions of ASHRAE 90.1 Appendix G required the energy modeler to determine Baseline parameters by referring to the prescriptive requirements in Sections 5 through 10 of the standard, ASHRAE 90.1-2016 Appendix G is self-contained, and includes the relevant referenced requirements within the Appendix. This should simplify the Baseline modeling process for projects using Appendix G.

Determine or update all relevant baseline inputs for the appropriate climate zone, building type, and building area.

When developing the baseline building performance model, ensure that additional HVAC system types in addition to the predominant HVAC system type are modeled as required in G3.1.1 (b) through (h). Spaces that are served by a different HVAC system in the proposed design due to load or schedule variances, different building functions, or cross-contamination requirements, will also often be served by a different system type in the baseline building due to the requirements stipulated in G3.1.1(b) through (h). Examples include:

- Per G3.1.1(b), a security office operating 24x7 and a kitchen with high peak summer cooling loads located in a midrise office building will each be modeled with a single zone system in the Baseline due to peak thermal loads that differ by 10 Btu/h*ft2 or more from the average of other spaces served by the system, and/or schedules that differ by more than 40 equivalent full-load hours from other spaces served by the system.
- Laboratory spaces in buildings with significant laboratory exhaust will be modeled as a single VAV system serving only those spaces per G3.1.1(d).
- A mixed use residential and non-residential building will be modeled with both residential and non-residential system types if the total area associated with each space type exceeds 20,000 square feet per G3.1.1(b).
- A heated only warehouse space will be modeled with a heated-only system (System type 9 or 10) while the adjacent office area will be modeled with both cooling and heating per G3.1.1(e).

If the energy simulation software automates some or all the baseline generation, review the automated baseline model inputs against the expected baseline values and confirm consistency (see *Further Explanation, Common Issues with Energy Modeling*).

Preparation of the initial baseline building performance model is best undertaken during the design development phase, after major design decisions have been made, so that modeling can evaluate whether the project is likely to meet energy savings targets (or achieve points under the related credit). The baseline building performance model will typically need to be updated upon completion of the final project design.

Schedules

If anticipated operating schedules are unknown, helpful guidance for determining model inputs for occupancy, lighting, HVAC system, receptacle power, and service hot water consumption values can be found in the ASHRAE 90.1–2016 User's Manual, Appendix G.

Align the time steps used in the modeled schedules with the time steps used for determining peak demand. For elevators, the modeled peak power for the elevator motors must either be scaled down from the instantaneous peak by an equal factor in the baseline and proposed design to represent the hourly peak, or the fractional schedule modeled for elevators shall have a peak operating percentage that aligns with the peak hourly demand as a fraction of the peak instantaneous demand. For example, if the peak elevator motor power for the proposed design is 100 kW, the peak elevator motor power for the baseline design is 120 kW, and the peak hourly demand for the proposed elevators is 5 kW, the elevator motor power shall either be modeled as: hourly schedules that peak at 100% of design power, and peak hourly demand modeled as 5 kW in the Proposed Design and 6 kW in the Baseline design; or hourly schedules that peak at 5% (including for Baseline HVAC system sizing), and peak hourly demand modeled as 100 kW in the Proposed Design and 120 kW in the Baseline design.

Similar schedule or peak hourly load adjustments apply for receptacle power (with maximum connected load versus peak hourly demand) and service water heating (with maximum instantaneous flow versus peak hourly demand).

Schedules must be identical in both the baseline and the proposed cases unless documented in an exceptional calculation or specifically allowed by ASHRAE 90.1–2016 Appendix G (see *Further Explanation, Exceptional Calculation Method*).

Certain space types may require specific schedules based on anticipated operation and may vary by space type. For example, a server room may have different temperature schedules than an occupied space. Exceptions to Section G3.1.1 may require modeling of a different baseline HVAC system type in spaces with schedules that vary significantly from the rest of the building.

Different lighting schedules should be used for a project with both office and retail occupancy when the space-by-space method is used, or when the building area method is used with multiple building type classifications. Different schedules cannot be used, however, if an average lighting power density is applied to the whole project.

Energy Cost

For EA Prerequisite Minimum Energy Performance and EA Credit Optimize Energy Performance, baseline and proposed costs must be determined on an energy cost basis using actual utility rates or the state's average energy prices referencing US Energy Information Administration (EIA) state average annual rates per fuel source for the most recent year for projects in the U.S., or the most recent published national or provincial average annual rates per fuel source for projects outside the U.S.

For each energy source serving the building, the utility rates or tariffs must be identical for the Baseline and Proposed building models.

For projects using actual rates:

- The modeled rates shall include all components of the rates applicable to the building (i.e. consumption charges, demand charges, monthly or time-of-use variations in rates, distribution and transportation charges, etc.). Monthly or average annual rates shall be limited to applications where this is the only information available to the project (i.e. campus projects where blended rates are calculated for the whole campus).
- If the baseline model has a fuel source not included in the proposed design (i.e. natural gas or propane), the rates shall be determined based on current rates available in the project location (e.g. a current natural gas tariff applicable to similar buildings that have natural gas service in the project location).

Energy cost savings is referenced as a metric for overall building energy efficiency:

- It aligns with the energy modeling procedures in ASHRAE 90.1–2016, Appendix G
- It captures the relative effects of various efficiency measures on energy demand and long-term operating costs—valuable metrics for the owner in determining the overall cost-effectiveness of selected efficiency strategies.
- It can help designers understand energy consumption because in many cases, cost and environmental impacts of each fuel source are correlated.

For the cost metric, on-site renewable energy that complies with ASHRAE 90.1-2016 Section G2.4.1 requirements may be documented for credit under EA Prerequisite Minimum Energy Performance and EA Credit Optimize Energy Performance. Off-site renewable energy and any on-site renewable energy not meeting the criteria of Section G2.4.1 is ineligible for credit when applying the cost metric (See Further Explanation, Applying Renewable Energy Savings).

Greenhouse Gas Emissions

LEED v4.1 incorporates greenhouse gas emissions (also referred to as CO2 equivalent emissions) as a metric for building energy performance. Understanding greenhouse gas emissions from building energy use and prioritizing building emissions reductions is critical for addressing climate change.

The total greenhouse gas emissions metric, in terms of carbon dioxide equivalents, shall be calculated for the baseline building performance rating and for the proposed building performance rating, and the percentage improvement shall be determined using carbon dioxide equivalent emissions. For each energy source serving the building, the GHG emission factors must be identical for the Baseline and Proposed building models.

Greenhouse gas emissions factors per fuel source shall be determined as follows (the most recently available annual data shall be used unless otherwise indicated):

- United States or Canada:
  - Use U.S. Environmental Protection Agency's (EPA) Greenhouse Gas Factors for the most recently available year (see Energy Star Portfolio Manager Technical Reference: Greenhouse Gas Emissions or https://www.epa.gov/egrid/data-explorer).
    - For electricity, use indirect greenhouse gas emission factors per eGRID region in the U.S., and indirect greenhouse gas emission factors per province in Canada.
    - For fuel (including biomass), use direct GHG emissions factors per fuel source, as applicable.
    - For purchased heating or purchased chilled water, use indirect GHG emissions factors per fuel source.

    OR

  - Continental US (required if documenting Optimize Energy Performance Credit greenhouse gas emissions savings for Tier 2 new off-site renewables)
    - Electricity hourly emissions factors: Use hourly Cambium Levelized Long-Run Marginal Emission Rates published by NREL for the project's Generation and Emission Assessment (GEA) region with characteristics as defined in the *LEED Hourly Cambium, Demand Adjusted Energy Metrics Calculator*. The hourly electricity consumption is multiplied by the hourly CO_2e emissions factor to determine the hourly CO_2e emissions.
    - Other Fuels: Use the total CO_2e emissions factors that include both direct and indirect CO_2e referenced in the *LEED Hourly Cambium, Demand Adjusted Energy Metrics Calculator*.

    Note: *These references account for both indirect and direct emissions (i.e. combustion + pre-combustion), leading to higher reported GHG emissions for fossil fuel and for the portion of eGRID electricity generated by fossil fuel).*

- European Union:
  - European Environment Agency (EEA) – National emissions reported to the United Nations Framework Convention on Climate Change (UNFCCC) and complete energy balances from Eurostat. (https://www.eea.europa.eu).

- All other locations, and EU countries without EEA data reporting
  - Climate Transparency - Enerdata national emissions factors (https://www.climate-transparency.org; https://www.enerdata.net/).

For references above that report only electricity emissions factors, use the U.S. Environmental Protection Agency's (EPA) direct GHG emissions per fuel source published in the Energy Star Portfolio Manager Technical Reference: Greenhouse Gas Emissions for all non-electric fuel sources.

Refer to *Further Explanation, Applying renewable savings* for further guidance on demonstrating renewable energy savings within the energy model. For projects in the continental United States, note that the contract for Tier 2 New Off-site renewable energy must include sufficient information to identify the grid region, and a reasonable estimation of the hourly generation profile to contribute to GHG emissions savings in EAc Optimize Energy Performance.

Custom GHG Emissions factors:

Projects outside the U.S. may document custom GHG Emissions factors to be approved on a case-by-case basis for published national or regional CO2 equivalent emissions per fuel source, particularly references that include hourly electric emissions factors. Supplemental documentation shall be provided to demonstrate that the methodology for calculating the emissions factors, and assumptions per fuel input source are derived consistently. For reporting methods that separate "renewable" and "non-renewable" emissions (e.g. for wood fuel classified as renewable in the project location; or for national average electric emissions that separately report renewable versus non-renewable emissions), the total renewable plus non-renewable emissions factors shall be used. For non-electric fuel sources, if the electricity emissions factors are calculated based on fuel inputs that include both indirect and direct emissions (i.e. combustion + pre-combustion), use the total CO2e emissions factors that include both direct and indirect CO2e referenced in the *LEED Hourly Cambium, Demand Adjusted Energy Metrics Calculator*.

Electricity hourly emissions factors:

For a more granular picture of greenhouse gas emissions reduced through building efficiency, renewables procurement, and grid harmonization strategies, projects are encouraged to use hourly electricity emissions profiles when available. In the continental United States, NREL has published Cambium hourly long run marginal emissions profiles estimated based on prospective power sector scenarios for each Generation and Emission Assessment (GEA) region. The specific power sector scenario and characteristics used for LEED documentation are referenced in the *LEED Hourly Cambium, Demand Adjusted Energy Metrics Calculator*.

Hourly emissions for baseline electricity, baseline unregulated electricity, and proposed electricity are calculated by multiplying the hourly electricity greenhouse gas emission factors from the project's grid region by the hourly electric consumption from the energy simulation for the baseline design, baseline design unregulated electricity, and proposed design respectively. The hourly electric emissions for each case are summed together to determine the associated annual greenhouse gas emissions. Hourly emissions for qualifying Tier 2 off-site renewable energy are determined based on the renewable asset's GEA grid region and hourly electricity generation from the renewable asset (See Further Explanation, Applying renewable savings).

Some energy software can include this hourly emissions calculation directly into the energy model (similar to a utility rate), while other energy software may require post-processing of hourly electric consumption to perform the simple calculation. For projects in the continental United States, the *LEED Hourly Cambium, Demand Adjusted Energy Metrics Calculator* completes this post-processing based on hourly electricity data and monthly non-electric energy data imported into the calculator from the energy model. Most energy simulation software provides hourly and monthly energy outputs in a delimited format and can be quickly imported into the calculator.

For prerequisite compliance, only renewable energy meeting the ASHRAE 90.1-2016 definition for on-site renewable energy may be modeled for credit when using the GHG metric for compliance. For Optimize Energy Performance credit compliance, Tier 2 off-site renewable energy may also be modeled for credit. For projects claiming credit for GHG emissions reductions associated with Tier 2 off-site renewable energy in EA credit Optimize Energy Performance, building GHG emissions and avoided GHG emissions from Tier 2 off-site renewable energy can be modeled using the hourly generation profile from the project's grid region and the renewable generation grid region in locations where hourly emissions factors are available.

Applying Renewable Energy Savings

Credit for on-site and new off-site renewable energy may be documented as follows, either directly in the proposed energy model, or applying post-processing of the proposed energy model results.

- On-site renewable energy:
  Energy costs and greenhouse gas emissions savings associated with on-site or on-campus renewable energy systems may be documented for EA Prerequisite Minimum Energy and EA Credit Optimize Energy Performance when the following criteria are met:

- Per ASHRAE 90.1-2016 Section G2.4.1, the renewable systems are included on the building permit and used in the building. LEED also recognizes renewable systems that are on a master site permit for a contiguous campus that includes the LEED project when the renewable energy is used in the contiguous campus.
- Energy is generated from renewable sources produced at the building site or master site per ASHRAE 90.1-2016 definitions.
- All associated environmental attributes are retained by the building owner.

Note: earlier versions of LEED allowed some biofuels produced off-site to qualify as on-site renewable energy. However, based on the clarifications provided in ASHRAE 90.1-2016 for on-site renewable energy, and the clearer distinction between on-site and off-site renewable energy in LEED v4.1, biofuels are only considered on-site renewable systems when the renewable source is harvested on site or on a contiguous campus, and used for on-site generation of electric or thermal energy. Furthermore, the renewable system must be part of the LEED project scope of work (or campus development scope of work including the project) under the ASHRAE 90.1-2016 requirements. Therefore, electric generation or thermal generation from renewable fuels sourced off-site and transported to the building site are not considered on-site renewable generation.

The renewable energy generation shall be documented using a method that accounts for all losses, with relevant design criteria and local climatic conditions incorporated into the calculations. Free web-based tools produced by U.S. national laboratories are available for estimating annual, monthly, and hourly energy generation from solar photovoltaics. For on-site renewable generation of electricity, projects using net metering may claim credit for all electric generation on an annual basis, up to 100% of the electricity documented in the energy model (including electricity associated with District Energy Systems serving the project, and documented using v4 Reference Guide, *Further Explanation, Project Type Variations, District Energy Systems, Full DES Performance Accounting*). For on-site renewable generation of thermal energy, documentation shall be provided to support that the hourly thermal energy generation, available thermal storage capacity, and relevant thermal losses are evaluated in conjunction with the hourly thermal load profile.

ASHRAE 90.1-2016 Section G2.4 states that on-site renewable energy "shall be subtracted from the *proposed design energy* consumption prior to calculating the *proposed building performance.*"

- Cost Metric: The equivalent cost of the on-site renewable energy system can be calculated in two ways, virtual rate or actual utility tariff.
  - Virtual rate. The project team may use the virtual energy rate determined by the proposed building energy model used for EA Credit Optimize Energy Performance. The virtual rate accounts for both consumption and demand charges. Project teams that use the Energy Information Administration's average energy prices must use the virtual rates to determine the renewable energy system cost.
  - Actual utility tariff. Calculate the expected savings in both consumption and demand charges, based on the rates charged by the utility that serves the project. If a project is served by a utility that uses time-dependent valuation to set rates, the team must provide hourly calculations for the value of generated energy. Some energy modeling software may calculate the savings from renewable energy systems if the utility rates include consumption, demand, time-dependent valuation, time-of-use, ratchets, and other factors.

Annual cost savings for renewable electricity generation shall not exceed 100% of the annual electricity cost documented in the proposed energy model before applying renewable savings.

- Greenhouse gas emissions metric: the proposed emissions associated with the on-site renewable generation shall be calculated using the same methodology as the proposed design (i.e. annual average GHG emissions factors per energy source, or hourly electric emissions). If using hourly electric emissions, the project must document the hourly renewable generation profiles for any on-site renewable electric generation. Annual GHG emissions savings for on-site renewable electric generation on an annual basis shall not exceed 100% of the annual electric GHG emissions documented in the energy model before applying renewable savings.

- Tier 2 New Off-site renewable energy:
  Greenhouse gas emissions offset by Tier 2 New off-site renewable energy systems qualifying under EA credit Renewable Energy may be included in the model for achievement of points using the greenhouse gas emissions calculation under EA credit Optimize Energy Performance, but may not be included in the model for EA Prerequisite Minimum Energy Performance compliance.

  <u>Hourly GHG Emission requirements:</u>
  Projects located in the continental United States or in any other location where hourly GHG electric greenhouse gas emissions factors are available must document performance for the baseline, proposed design, and renewable generation using hourly electric emissions factors. Therefore, for projects in the continental United States, the contract for the renewable asset must include sufficient information to identify the grid region, and a reasonable estimation of the hourly generation profile for the renewable asset. Many larger contracts include the renewable project location and renewable generation hourly profile in the contract documents, as a set of twelve monthly profiles, indicating the typical 24-hour generation profile for each month of the year. For smaller contracts, if the renewable asset location and characteristics are indicated on the contract, the hourly generation profile can be estimated:
  - Photovoltaic generation: If the photovoltaic generation location, module type, array type, system losses, tilt, and azimuth are known, NREL's free online PVWatts tool may be used to calculate the hourly solar generation profile.
  - Wind generation: If the wind generation location and annual generation amount are known, hourly wind generation data from the US Hourly electricity grid monitor for the balancing authority closest to the project generator may be used to generate an hourly profile, calculating the hourly wind generation as a percentage of total annual wind generation for that location (https://www.eia.gov/electricity/gridmonitor/).

  Avoided hourly GHG emissions from Tier 2 New off-site renewable energy are calculated by multiplying the hourly electricity greenhouse gas emission factors from the renewable asset's grid region by the hourly electric generation from the renewable energy system. For projects in the continental United States, these hourly calculations are performed automatically in the *LEED Hourly Cambium, Demand Adjusted Energy Metrics Calculator* based on data inputs indicating the hourly renewable generation and the renewable asset's grid region. Annual GHG emissions savings for renewable electric generation shall not exceed 100% of the annual electric GHG emissions documented in the proposed energy model before applying renewable savings.

- Tier 1 on-site renewable energy systems that do not meet the contractual requirements of ASHRAE 90.1-2016 Section G2.4 may document achievement of points using the greenhouse gas emissions calculation under EA credit Optimize Energy Performance using the guidance for Tier 2 New off-site renewable energy systems.

Exceptional Calculation Method

Refer to the LEED v4 reference guide*,* with the following modifications:

- Replace all instances of ASHRAE 90.1–2010 with ASHRAE 90.1–2016
- Delete the first paragraph in the *Additional guidance* section
- Delete the *Changes from earlier versions of ASHRAE and LEED* section
- Refer to Rating System Variations, Data Centers for exceptional calculations that may be applied to data centers that comprise a portion of the project area.
- Refer to Rating System Variations, Retail for exceptional calculations that may be applied to commercial cooking equipment, appliances, or commercial refrigeration.

ASHRAE 90.1-2016 versus 2010

The referenced standard for building the baseline model for this prerequisite has been updated to ASHRAE 90.1–2016, which represents a substantial increase in efficiency from ASHRAE 90.1–2010. Some of the major changes are described in Tables 1 and 2. Detailed changes between ASHRAE 90.1-2010 and ASHRAE 90.1-2013 are described in ASHRAE 90.1-2013 Appendix F. Detailed changes between ASHRAE 90.1-2013 and 90.1-2016 are summarized in ASHRAE 90.1-2016 Informative Appendix H.

Table 1. Changes in ASHRAE 90.1 mandatory requirements, 2010 to 2016

Building Envelope Requirement	ASHRAE 90.1-2016
Heated or Cooled Vestibule requirement (6.4.3.9)	Requires heated or cooled vestibules to limit setpoint temperatures, and automatically shut off heating when outdoor air temperature exceeds a certain level.
Verification of Envelope requirements (4.2.4, 4.2.5, 5.2.1, 5.2.9)	Adds verification requirements for envelope components including insulation, air leakage, and other properties.
HVAC & Refrigeration Requirement	**ASHRAE 90.1-2016**
Refrigerators and Freezers (Tables 6.8.1-12 and 6.8.1-13)	Maximum energy consumption regulated for some commercial refrigerators and freezers
HVAC Equipment (Tables 6.8)	Increased efficiencies for HVAC equipment, and increased capacity control for some packaged equipment
Humidification and Dehumidification (6.4.3.6)	Increased control requirements prohibiting the use of fossil fuel and electricity for humidification above 30% RH and dehumidification below 60% RH in most circumstances.
Demand Control Ventilation (6.4.3.8)	Reduces the occupancy threshold where DCV is required from 40 people per 1,000 ft^2 to 25 people per 1,000 ft^2
Heating and Cooling Setbacks (6.4.3.3)	Requires heating setback at least 10°F (6°C) below occupied heating setpoint, and cooling setback at least 5°F (3°C) above occupied cooling setback. *Note: These setbacks must be part of the Baseline and Proposed schedules modeled using the Performance method.*
Optimum start control (6.4.3.3.3)	Optimum start controls required for more building types
Duct Insulation (6.4.4.1.2)	Increases ductwork insulation requirements

DDC control (6.4.3.10)	DDC Control required for a much larger array of building applications
HVAC alterations (6.1.1.3.1)	Requires replacement HVAC&R equipment to meet most requirements
Pool Dehumidifier (6.4.1.1)	Establishes efficiency requirements for indoor pool dehumidifier
Fault Detection (6.4.3, 6.4.3.12)	Adds fault detection requirements
Power Requirement	**ASHRAE 90.1-2016**
Automated receptacle control (8.4.2)	Expands the spaces where automated receptacle control is required and provides further details regarding acceptable methods for receptacle controls
Electrical Monitoring (8.4.3)	Adds monitoring requirements to submeter tenant energy and electric end uses
Automated receptacle control (8.4.4)	Adds transformer performance requirements
Lighting Requirement	**ASHRAE 90.1-2016**
Daylighting Controls (9.4.1.1)	Requirements updated for areas where mandatory daylighting controls are required.
Automatic shutoff of lighting and switched receptacles in hotel guestrooms (9.4.1.3)	Adds requirements for automated shutoff of lights and switched receptacles in hotel/motel guestrooms
Lighting Controls (9.4.1.1)	Additional lighting controls requirements including partial automatic ON, inclusion of emergency circuits in scheduled shutoff requirements, additional shutoff controls for exterior lighting, increased parking garage occupancy controls.
Lighting Efficacy (9.4.1)	Adds efficacy requirements for residential dwelling unit lighting
Lighting Alterations (9.1.2)	Increases requirements for alterations to existing building lighting systems
Exterior Lighting Power (9.4.2)	Reduces exterior lighting power allowances
Motor Requirement	**ASHRAE 90.1-2016**
Motor Efficiency (10.4.1)	Increases motor efficiencies
Escalators (10.4.3 and 10.4.4)	Adds requirements for escalators, moving walkways, and elevators
Whole Building energy monitoring (10.4.5)	Adds requirement to monitor whole building energy use for energy supplied by a utility, energy provider, or plant not located in the building

Table 2. Changes in ASHRAE 90.1 prescriptive requirements, 2010 to 2016

Building Envelope Requirement	ASHRAE 90.1-2016
Opaque and Fenestration Efficiencies (Tables 5.5-1 through 5.5-8)	More stringent insulation levels for opaque elements in most climate zones. Fenestration: More stringent U-factor requirements for most assemblies, more stringent SHGCs in warmer climates. Additional fenestration framing types added.
Fenestration area by orientation (5.5.4.5)	Specific limitations added for fenestration area by orientation

Fenestration Visible Transmittance (5.5.4.6)	Minimum visible transmittance to solar heat gain coefficient ratio added.
HVAC & Refrigeration Requirement	**ASHRAE 90.1-2016**
Heat Rejection Fan Control (6.5.5.2)	Fan control required for multi-cell heat rejection equipment
Cooling tower flow turndown (6.5.5.4)	Cooling towers with multiple or variable speed condenser water pumps have added controls requirements associated with flow rate
Small motors (6.5.3.5)	Most motors under 1 hp required to be electrically commutated or have minimum efficiency of 70%.
Boiler Turndown (6.5.4.6)	Large boilers required to have minimum turndown ratio
Fan Power Allowance (Table 6.5.3.1B)	Changes to fan power pressure adjustments. Some allowances previously allowed to be used for a broad range of systems such as fully ducted return and exhaust are limited to specific systems.
Dehumidification (6.5.2.3)	Requires most reheat used for dehumidification to be from recovered or site-generated sources
Fluid Flow (6.5.4.1 through 6.5.4.3)	Requires automatic shutoff of pumps and boilers when fluid flow through the chillers or boilers is not operating, reduces low flow limit exceptions, requires variable flow in more hydronic system applications
Computer rooms (6, 6.6)	Adds requirements specific to computer rooms, including air and water economizer requirements
Transfer air (6.5.7.1)	Limits conditioned supply of transfer air between spaces
VFD Return and Relief Fans (6.5.4.1, 6.5.4.3)	Requires VFD control of return and relief fans larger than 0.5 hp.
Fan Powered VAV control	Specifics control of fans in fan-powered parallel VAV boxes
Energy Recovery (Tables 6.5.6.1-1 and 6.5.5.6.1-2)	Revises minimum threshold for energy recovery
Water-side economizers	Requires water-side economizers for radiant cooling or passive chilled beam systems
Lighting Requirement	**ASHRAE 90.1-2016**
Interior Lighting Power Density (Tables 9.5.1 and 9.6.1)	Extensive changes to the Interior Lighting Power Density requirements.
Decorative Lighting (9.6.2)	Reduces additional lighting allowance for decorative lighting

Table 3. Changes in ASHRAE 90.1 Performance Rating Method Requirements, 2010 to 2016

General Requirement	ASHRAE 90.1-2016 Appendix G
Performance Rating Method Scope (G1.1)	Appendix G can be used to demonstrate code compliance (Previously it only applied to demonstrate above-code performance).
Performance Rating Calculation (G1.2.2 / 4.2.1.1)	A stable baseline that references ASHRAE 90.1-2004 prescriptive values is introduced. Performance Cost Index Target (PCI_T) is calculated using Building Performance Factors (BPF) for each building type and climate zone in conjunction with Baseline Building Unregulated Energy Consumption (BBUEC) and Baseline Building Performance (BBP).

	Building Performance Factors represent the ratio of regulated energy cost for a 90.1-2016 versus a 90.1-2004 compliant building. This allows relatively few changes to the Baseline Building modeling methodology between code cycles, with the major change being the BPF determinations.
Self-contained references	The Baseline modeling requirements are contained within ASHRAE 90.1-2016 Appendix G, and do not require references to the prescriptive requirements of Sections 5 through 10.
Unmodified existing building components (Table G3.1#2(Baseline))	Unmodified existing building components are required to follow the same rules as new and modified building components. Previously some existing building components (such as existing building envelope components) could be modeled using existing unrenovated performance in the Baseline and as-designed with renovations in the Proposed.
Unfinished spaces	For unfinished spaces, the proposed efficiencies, controls, lighting power densities, etc. are modeled consistent with the ASHRAE 90.1-2016 prescriptive requirements, and are not modeled identically to the Baseline.
Schedules	ASHRAE 90.1-2016 Appendix G
HVAC setpoint schedules (Table G3.1#4)	Projects may adjust schedules to demonstrate credit for HVAC systems that automatically provide occupant thermal comfort via means other than direct control of air dry-bulb and wet-bulb temperature.
Building Envelope Requirement	ASHRAE 90.1-2016 Appendix G
Vertical Fenestration Area (Table G3.1#5(Baseline)(c))	Vertical fenestration area modeled in the Baseline is less than 40% for many building occupancies. Credit is allowed when the proposed vertical fenestration area is lower than the values shown in Table G3.1.1-1 for applicable building types.
Infiltration (Table G3.1#5(Proposed)(b), G3.1.1.4)	Specific infiltration rates are required to be modeled. Credit is allowed for improved infiltration for projects performing air leakage testing.
HVAC & Refrigeration Requirement	ASHRAE 90.1-2016 Appendix G
HVAC System Type (G3.1.1, Table G3.1#10, Table G3.1.1-3, Table G3.1.1-4)	Baseline system heating type is dependent on climate zone rather than predominant heating source used in the building. Baseline systems in climate zone 0 to 3A are modeled with electric heating and baseline systems in climate zones 3B through 8 are modeled with fossil fuel heating. Baseline fossil fuel heating systems shall always be modeled using natural gas, or propane in locations where natural gas is not available. Previous versions of Appendix G required the fuel type to be the same in the Baseline and Proposed case.

	Further clarity is provided for identifying the order of priority for determining the Baseline HVAC system types applicable for the building. Additional HVAC system type categorizations added for public assembly, retail buildings up to two floors, hospitals, computer rooms.
HVAC equipment efficiencies (G3.1.2.1)	Projects are required to model both part load and full load efficiencies per Tables G3.5.1 through G3.5.6 where applicable. A clear method is provided for calculating the modeled Baseline cooling and heating COP for packaged equipment.
Night-time fan cycling (G3.1.2.4)	For System 6 and 8 (Parallel fan-powered VAV terminals with electric heating), the terminal unit fan and reheat coil are energized to meet the heating unoccupied setpoint in the space rather than the entire VAV system serving the floor.
Computer room fluid economizer (G3.1.2.6.1)	Computer room fluid economizers required for computer rooms where the Baseline system type is system 11.
Baseline Humidity Controls (G3.1.3.18 and Table G3.1#10(Baseline)	If the Baseline system type does not comply with humidistatic control requirements, then only 25% of system reheat energy shall be included in the baseline building performance. If the proposed design includes humidification, the baseline design shall use adiabatic humidification
Baseline Preheat (G3.1.3.19)	Preheat is required to be modeled for Baseline Systems 5 through 8, and controlled to a fixed setpoint 20°F (11°C) less than the design room heating temperature set point. Modeling of preheat in the Baseline is no longer dependent on the presence of preheat in the Proposed design.
Baseline Refrigeration (Table G3.1#17)	Refrigeration equipment is required to be modeled as specified.
Lighting Requirement	**ASHRAE 90.1-2016 Appendix G**
LPD Modeling Method (Table G3.1#6(Baseline))	All building spaces are modeled using the Space-by-Space Method. (Previous versions of ASHRAE allowed projects to use the space-by-space or building-area methods).
Automated Controls (Table G3.1#6(Proposed))	Control credit is modeled in the Proposed design for all spaces where applicable controls are included (including mandatory controls required by 90.1-2016).
Service Water Heating	**ASHRAE 90.1-2016 Appendix G**
Baseline System Type (Table G3.1.1-2)	Baseline service water heating system type is determined based on building type, with electric resistance point-of-use for convenience store, electric resistance storage for most commercial applications with low service water heating usage, and gas storage water heaters for residential buildings and commercial applications with high service water heating usage. In previous versions of ASHRAE, the service water heating type was modeled identically in the Baseline and Proposed Case
Service Water Heating Loads (Table G3.1#11(Baseline)(h)	A specific methodology is used for determining service water heating loads. Loads must be modeled identically in the baseline and proposed case, except when calculations show savings associated

	with reduced fixture flow, reduced required temperature of service mixed water, heat recovery for makeup water, etc.
Power and Equipment Requirement	**ASHRAE 90.1-2016**
Computer room equipment schedule (G3.1.3.16)	The computer room equipment schedule is varied monthly between 25% and 100% of full load as noted.
Elevators (G3.9.2, Table G3.1#16)	A specific methodology is provided for calculating baseline and proposed annual elevator energy consumption, Baseline elevator peak motor power, baseline elevator cab ventilation, and baseline elevator lighting power density.

Additional Energy Modeling Guidance

Thoroughly review both ASHRAE 90.1–2016 and the 90.1–2016 User's Manual. The manual presents extended explanations and also includes examples of the concepts and requirements within the standard.

The Pacific Northwest National Laboratory (PNNL) ANSI/ASHRAE/IES Standard 90.1-2016 Performance Rating Method Reference Manual also provides detailed modeling guidance which can be used when developing a 90.1-2016 Baseline and Proposed model (https://www.pnnl.gov/main/publications/external/technical_reports/PNNL-26917.pdf).

Rating System Variations

Core and Shell

Refer to the LEED v4 reference guide, with the following modifications:

- Replace instances of ASHRAE 90.1–2010 with ASHRAE 90.1–2016
- Energy cost savings are based on a building's total annual energy consumption, rather than on the owner's scope of work, so the owner of a core and shell project may have only a limited opportunity to improve energy savings. The percentage improvement thresholds for achievement of Option 1 in EA credit Optimize Energy Performance are therefore lower for Core and Shell projects than for New Construction.
- For unfinished spaces, the proposed efficiencies, controls, lighting power densities, etc. are modeled consistent with the ASHRAE 90.1-2016 prescriptive requirements, and are not modeled identically to the Baseline.

Retail

For all process loads, define a clear baseline for comparison with the proposed improvements. the baseline and design as follows:

- Commercial kitchen equipment addressed by California EnergyWise energy cost calculators may use the energy savings and percentage annual cost savings documented in these calculators to determine the Baseline annual energy consumption and Proposed annual energy consumption associated with the equipment, without additional documentation. Commercial kitchen equipment that has California EnergyWise options for natural gas or electricity may optionally be modeled using natural gas in the baseline and electricity in the proposed design, or may be modeled using the same energy source in the baseline and proposed.
- *Appliances and equipment.* For appliances and equipment not covered by California EnergyWise energy cost calculators, indicate hourly energy use for proposed and budget equipment, along with estimated daily use hours. Use the total estimated appliance/equipment energy use in the energy simulation model as a plug load. Reduced use time (schedule change) is not a category of energy improvement in this credit. ENERGY STAR ratings and evaluations are a valid basis for performing this calculation.
- *Display lighting.* For display lighting, use the space-by-space method of determining allowed lighting power under ANSI/ASHRAE/IESNA Standard 90.1–2016, with errata (or a USGBC-

approved equivalent standard for projects outside the U.S.), to determine the appropriate baseline for both the general building space and the display lighting.
- *Refrigeration.* For hard-wired refrigeration loads, model the effect of energy performance improvements with a simulation program designed to account for refrigeration equipment.

Data Centers
Refer to the LEED v4 reference guide, with the following modifications:
- Replace the *Modeling requirements* section with the section below.
- The computer room equipment schedule is varied monthly between 25% and 100% of full load as noted in ASHRAE 90.1-2016.

Modeling requirements
Energy modeling is required for all data center projects.

IT equipment energy and electrical infrastructure energy savings
Because of the high process loads associated with IT equipment and its electrical infrastructure, many project teams look to these traditionally unregulated energy end uses for energy savings. Though not required, if the project team is attempting to claim energy savings from these end uses, the data center calculator may provide a simplified method (see *Data Center Calculator*, below).

The reduced energy consumption of the IT and electrical equipment can help reduce HVAC energy usage. Project teams have the option of claiming the process load savings in isolation or creating an additional energy model based on the adjusted loads to capture the associated HVAC energy savings.

To determine total energy cost savings, it may be necessary to create three energy models. Below is a list of the models that may need to be created. The specific requirements of each model are detailed below.

1. Proposed model with as-designed IT loading (normal performance rating method, PRM, model)
2. ASHRAE baseline model with as-designed IT loading (normal PRM model)
3. ASHRAE baseline model with “baseline” IT loading (optional)

If the project team is claiming energy savings related to the IT systems, the total energy savings are calculated between models 1 and 3.

Proposed model with as-designed IT loading (model 1)
The model of the building’s energy cost must include all regulated energy end uses as listed in the prerequisite criteria, as well as any unregulated energy that is building-specific. The proposed design must use the IT loads and developed for the project and the schedule stipulated in ASHRAE 90.1-2016. The IT loads should be at the values for the intended final buildout of the facility. All electrical system components—examples include incoming transformers, switchgear, UPS systems, and power distribution units—must be modeled. Power losses associated with this equipment should be assigned to the spaces that house the equipment as an electrical load and as a thermal load input to the energy model. Model the quantity of power and cooling equipment designed to run during normal operation to include the effects of operating redundant equipment at partial loading on energy use.

In addition to the ASHRAE 90.1 mandatory compliance requirements, provide energy efficiency data for the following items:

- Generator block heaters (wattage required to keep the block at the design temperature)
- Power distribution wiring
- Battery charging

Submit documentation for the following items, showing efficiency data at initial and full system loading points (loading values are a percentage of total IT load):

- Service transformers
- Switchgear

Uninterruptible power systems

- Power distribution units

ASHRAE model with full IT loading (model 2)
Model using the same IT load as the Proposed design, with Baseline inputs consistent with ASHRAE 90.1-2016 Performance Rating Method requirements.

ASHRAE Baseline model with "baseline" IT loading (model 3)
This model is used to calculate IT energy savings due to low-energy servers, virtualization, and efficient electrical system design. In contrast to the standard application of exceptional calculation methods to the proposed model, for data center projects, the exceptional calculation is applied to the baseline (model 3). Rather than reducing the energy used in the proposed design, the baseline is increased to reflect the energy usage typical of a data center.

For IT equipment, the USGBC data center calculator provides baseline documentation; if used, additional justification for the baseline IT loads is not necessary. IT equipment input is defined as the IT load as measured at the point of connection of the IT device to the electrical power system. IT equipment input captures the actual power load of the IT device exclusive of any power distribution losses and loads beyond IT devices, such as rack-mounted fans.

The losses associated with all UPS equipment, including that which serves mechanical equipment to achieve continuous cooling during a loss of power (e.g., pumps, air-handling units, and compressors), is considered not part of the IT energy usage but part of the energy consumption required to operate the data center.

If a hydronic cooling system is used for IT cabinets or computers, the energy consumed by the fans built into the cabinet and coolant distribution pumps should be considered HVAC energy use, not IT energy use.

Data Center Calculator

Refer to the LEED v4 reference guide.

Project Type Variations

<u>District Energy Systems</u>

Projects that are served by district energy systems (DES) may demonstrate compliance with EA Prerequisite Minimum Energy Performance and EA Credit Optimize Energy Performance by following one of the following methods.

- Path 1 ASHRAE 90.1-2016 Appendix G. No credit is documented for the purchased energy systems.
    - Path 1A. ASHRAE 90.1-2016 Appendix G.
    - Path 1B. ASHRAE 90.1-2016 Appendix G with ASHRAE 90.1-2022 Addendum a. (Revise the Appendix G methodology to remove the inherent penalty for DES)
- Path 2 Full DES performance accounting. Credit is documented for the purchased energy systems. The proposed design is modeled using a virtual plant consistent with the district energy system performance, and the baseline design is modeled with on-site systems from ASHRAE 90.1-2016 Appendix G for site generated thermal energy.
- Path 3 Large-scale District Energy Systems. GHG emissions savings associated with the upstream system are documented using the Large-Scale DES Calculator. No energy efficiency savings (using the cost or source energy metric) are documented for the upstream system.

The modeling path chosen by the project team may depend on the relative efficiency of the DES to which the project is connected, how much DES information is available, or whether an energy model already exists for the system. Whenever possible, incorporate system and equipment performance parameters directly into the energy simulation. Potential methods include developing efficiency curves and scheduling equipment operation and curves. Postprocessing of DES performance is acceptable if reasonable simulation methods are not available or are too onerous. All postprocessing methodologies must be fully documented.

All Paths: Scope of DES equipment inclusion
All downstream equipment must be included in the scope of EA Prerequisite Minimum Energy Performance and EA Credit Optimize Energy Performance. Downstream equipment includes heat exchangers, steam pressure reduction stations, pumps, valves, pipes, building electrical services, and controls.

Upstream equipment is included or excluded depending on the chosen path.

Path 1A. ASHRAE 90.1–2016, Appendix G
Model the proposed and baseline designs using purchased energy according to ASHRAE 90.1–2010, Appendix G.

Published purchased energy rates or conversion factors:
Energy Cost: Per ASHRAE 90.1, model the purchased energy rates for each district energy source (purchased hot water, purchased steam, or purchased chilled water) identically in the baseline and proposed design based on actual utility rates, if actual utility rates are available.

GHG Emissions: For the GHG emissions metric, if the published reference for electricity and fossil fuel GHG emission factors also reports emission factors for district energy (purchased hot water, purchased steam, or purchased chilled water), model these published emission factors for each district energy source identically in the baseline and proposed design.

Derivation of DES purchased energy rates or GHG emission factors when unpublished:
If purchased energy rates and/or GHG emission factors are not published for the district energy sources serving the project, derive these purchased energy rates and/or emission factors leveraging the electricity or fossil fuel data. For fossil fuel, use natural gas if the building does not receive fossil fuel and the district energy fuel source is unknown.

- District Chilled Water (CHW):
  CHWFactor = ElectricityFactor x 0.325
- District Hot Water (HHW):
  HHWFactor = FossilFuelFactor x 1.65
- District Steam Water (Steam):
  SteamFactor = FossilFuelFactor x 1.85

Where:

- For the Cost Metric (See further guidance below for purchased energy rates)
  - CHWFactor = Chilled water purchased energy rate ($ / unit energy)
  - HHWFactor = Hot water purchased energy rate ($ / unit energy)
  - SteamFactor = Steam purchased energy rate ($ / unit energy)
  - ElectricityFactor = Electricity purchased energy rate ($ / unit energy)
  - FossilFuelFactor = Fossil Fuel purchased energy rate ($ / unit energy)
    Units of energy must be consistent throughout each equation (i.e. consistently $/kWh or $/kBtu)
- For the GHG Emissions Metric:
  - CHWFactor = Chilled water GHG emissions factor

- HHWFactor = Hot water GHG emissions factor
- SteamFactor = Steam GHG emissions factor
- ElectricityFactor = Electricity GHG emissions factor
- FossilFuelFactor = Fossil Fuel GHG emissions factor
  Units of each GHG emissions factor must be consistent (in weight of CO2eq emissions per unit of energy)

Additional guidance: Cost Metric ElectricityFactor and FossilFuelFactor.
For the cost metric, in a flat rate structure, in which the building cost per unit of electricity or building cost per unit of natural gas is the same throughout the year and there are no demand charges, then those flat rates become the ElectricityFactor and FossilFuelFactor for the project cost metric. If all energy rate structures are not flat, a preliminary run of the Option 1 baseline case energy model must first be completed to identify the virtual electric rate (ElectricityFactor) and fossil fuel rate (FossilFuelFactor) for the project.

To obtain the virtual fuel rate (FossilFuelFactor) when the connected building does not use fossil fuel but the DES central plant does, use a flat rate consistent with the central plant rates or the historic average local market rates.

Path 1B. ASHRAE 90.1-2016 Appendix G with ASHRAE 90.1-2022 Addendum a.
The project may apply ASHRAE 90.1-2022 Addendum a to the ASHRAE 90.1-2016 Appendix G criteria. This eliminates the inherent penalty in the ASHRAE 90.1-2016 Appendix G Performance Index Targets when modeling "purchased heat" and "purchased chilled water". Replace all ASHRAE 90.1-2022 Addendum a references to Section 6 prescriptive criteria for the *proposed building design* with ASHRAE 90.1-2016 Section 6. For the *baseline building design* HVAC systems, the project team must exclusively reference ASHRAE 90.1-2022 with Addendum a. Free read-only versions of ASHRAE 90.1-2022 are available at https://www.ashrae.org/technical-resources/standards-and-guidelines/read-only-versions-of-ashrae-standards.

90.1-2022 Addendum a criteria as applied to ASHRAE 90.1-2016 Appendix G:

- Model HVAC systems for the *baseline building design* per ASHRAE 90.1-2022 Appendix G criteria as if all heating and cooling generation equipment is on-site;
- Projects with purchased heat: Model the *proposed building design* using natural gas forced draft boilers that prescriptively comply with ASHRAE 90.1-2016 Section 6 in lieu of purchased heat. The number of boilers and boiler controls shall meet the requirements of ASHRAE 90.1-2022 Section G3.2.3.2 through G3.2.3.6, without exceptions. Forced draft boiler efficiencies shall be modeled per the mandatory and prescriptive requirements of ASHRAE 90.1-2016 Section 6. Boiler systems with design input exceeding 1,000,000 Btu/h may document credit for minimum turndown ratios per Table 6.5.4.1.
- Projects with purchased chilled water: Model the *proposed building design* using water-cooled chillers that prescriptively comply with ASHRAE 90.1-2016 Section 6 in lieu of purchased chilled water.
  - Model the type and number of water-cooled electric chillers per ASHRAE 90.1-2022 Table G3.2.3.7 based on the peak coincident cooling load of baseline HVAC systems using chilled water (See 90.1-2022 Section G3.2.3.7).
  - Model the chilled water (CHW) with a design supply temperature of 44 °F (7 °C) and return temperature of 56 °F (13.3 °C) (See 90.1-2022 Section G3.2.3.8)
  - Model each chiller with separate condenser-water and primary chilled-water pumps interlocked to operate with the associated chiller per ASHRAE 90.1-2022 G3.2.3.11. Model the CHW loop as constant-flow primary and variable-flow secondary with the pump power of each loop modeled per 90.1-2022 Section G3.2.3.10, without exceptions. Model secondary loops with a pump motor demand of 30% of design wattage at 50% of design flow per the prescriptive criteria of ASHRAE 90.1-2016 Section 6.5.4.2. For

systems with total modeled chilled water capacity exceeding 300,000 Btu/h (25 kW) utilizing DDC CHW control valves, model chilled water supply temperature reset based on valve positions until one valve is wide open or setpoint limits have been reached per ASHRAE 90.1-2016 Section 6.5.4.4.

- Model heat rejection as an axial fan cooling tower with design fan power = 40.2 gpm/hp per ASHRAE 90.1-2016 Table 6.8.1-7, and with design supply temperature and leaving water temperature determined per ASHRAE 90.1-2022 G3.2.3.11. If the total fan power for the heat rejection equipment exceeds 5 hp, model the cooling tower fans with variable-speed fan controls that reduce fan motor demand to no more than 30% of design wattage at 50% of design air volume per ASHRAE 90.1-2016 Section 6.5.11.

Path 2. Full DES performance accounting

For path 2, the energy model scope accounts for both downstream equipment and upstream equipment and requires calculation of the district energy average efficiencies using engineering analysis or monitored data or a combination of both.

Energy rates (Cost Metric)

All DES electricity and fuel rates must be identical in both the baseline and the proposed cases. Use local electricity and fuel rates as they would normally apply to the building for the energy sources under consideration. If this information is not available, use representative market rates.

Exception: For District cooling or district heating plants without cogeneration or fuel cells that operate under specific and atypical electric rate structures and actively take advantage of those rates through strategies such as load management or energy storage, use the rate structures as they apply to the DES.

Greenhouse Gas Emissions Factors

See the guidance in Further Explanation, Greenhouse Gas Emissions.

Baseline building systems

For systems with thermal energy delivered from the district energy system, model the baseline case with on-site systems per ASHRAE 90.1-2022 Addendum a criteria described above.

Proposed building plant

Model the proposed case with a virtual DES-equivalent plant. Use the same efficiencies as the entire upstream DES heating, and cooling, and combined heat and power (CHP) systems, including all distribution losses and energy use.

Equipment efficiencies, distribution losses, and distribution pumping energy may be determined using any of the following methods:

- Monitored data
- Engineering analysis

Efficiencies and losses may be determined and modeled at any level of time resolution, from hourly to annual. However, the time resolution must be sufficiently granular to capture and reasonably represent any significant time- or load-dependent interactions between systems, such as thermal storage or CHP. Monitoring and analytical methods may be combined as necessary and appropriate. Monitoring data for heating, cooling, pumping, and cogeneration may be used only if the thermal loads that are monitored represent at least 90% of the load on the campus or district plant predicted after building occupancy. Whether using monitoring or an analytical method, the methodologies must be fully documented. The following specific requirements apply.

Heating and cooling plants
Efficiencies, whether determined through monitoring or analytically, must include all operational effects, such as standby, equipment cycling, partial-load operation, internal pumping, and thermal losses.

Thermal distribution losses
Use monitored data or an engineering analysis.

- Monitored data determine the distribution losses for the DES by measuring the total thermal energy leaving the plant and comparing it with the total thermal energy used by the buildings connected to the DES. Rate the plant efficiency accordingly in the energy model:

  Plant efficiency (%) x [100% – distribution loss (%)]
- An engineering analysis takes into consideration all distribution losses between the DES and the building. For distribution main losses, use a prorated amount based on load. For dedicated branch losses, use the total losses of the branch that feeds the building, including heat losses and steam trap losses. Compare the total losses with the total load of the building to get a percentage distribution loss relative to load and downgrade the plant's efficiency accordingly in the energy model.

  If thermal distribution losses are not measured or modeled, use the following default losses:

  - Chilled water district cooling, 5%
  - Hot water district heating, 10%
  - Closed-loop steam systems, 15%
  - Open-loop steam systems, 25%

    For steam systems that are partially open and partially closed, prorate between the above 15% and 25% losses in accordance with the fraction of expected or actual condensate loss.

Pumping energy
Whether through monitored data or engineering analysis, determine pumping energy for the project by prorating the total pump energy of the DES by the ratio of the annual thermal load of the building to the total annual DES thermal load. Model the pump energy as auxiliary electrical load. Pumping energy must be determined or estimated where it applies.

District Energy Combined Heat and Power (CHP)
To model the proposed design virtual plant, first monitor or model the total electricity generation, fuel input, and heat recovery associated with the District Energy Combined Heat and Power (CHP):

- Determine annual electricity generation using one of the following methods:
  - Monitor the total annual gross electricity generation. Also monitor the total annual parasitic loads, such as the annual electricity used for cooling the intake air for a turbine. Calculate the net annual electricity generation by subtracting all parasitic loads from the annual gross electricity generated.
  - Model the generators in energy simulation software per Appendix G. Use peak electricity efficiencies and generator curves that match the installed generators. Apply measured or estimated load profiles as process loads to reflect the estimated total electric and thermal loads on the district energy CHP system. Use the total energy generated and total fuel input from this analysis. Any parasitic loads must be included in the analysis and subtracted from the annual electricity generation.
- Calculate annual fuel input using one of the following methods:
  - Monitor the total annual fuel input to the generators.

- Model the generators in energy simulation software per Appendix G. Use peak electricity efficiencies and generator curves that match the installed generators.
- Calculate waste heat recovery using one of the following methods:
  - Monitor the total waste heat recovered.
  - Model the generators in energy simulation software per Appendix G. Use peak electricity efficiencies and generator curves that match the installed generators. Model the thermal equipment served by the CHP waste heat, such as boilers and absorption chillers, using the installed equipment capacities, efficiencies, and efficiency curves, and reflecting the total heating and cooling loads on the plant as a process load. Use the energy modeling outputs to identify the total heat recovered.

For baseline CHP electricity output, follow the general procedures described in this section for the proposed case, and adjust the results as follows depending on the results of the DES electricity allocation and the total modeled electricity use of the building in the Path 2 proposed case, including the electricity consumption of district plant equipment serving the building:

- Scenario A. If the building's allocation of CHP-generated electricity is less than or equal to its modeled electricity consumption, no adjustment is necessary. The baseline building is charged with the energy used by its (non-CHP) systems at market rates using standard procedures.
- Scenario B. If the building's allocation of CHP-generated electricity exceeds its modeled electricity consumption, include the amount of excess CHP electricity case as described in CHP fuel input formulas.

For the proposed design's CHP electricity output, allocate the electricity generation to the building based on the fraction of thermal loads to the building for the DES sources that use recovered waste heat. For each DES source supplied to the building, determine the fraction of the recovered waste heat applied to that source as well as the amount serving the project building. For relatively simple DES systems, in which the recovered waste heat is used directly in the DES, and for which waste heat serves only heating loads in the connected buildings, use the formula for simple systems:

$$CHP\_ELEC_{BLDG}(\text{simple systems}) = (X_{HEAT} \times BLDG_{HEAT}) \times CHP\_ELEC_{TOTAL}$$

where

CHP_ELEC_{BLDG}	=	CHP electricity generation allocated to building
X_{HEAT}	=	fraction of CHP plant's total production of waste heat applied to the DES directly
$BLDG_{HEAT}$	=	fraction of total district heat provided to building
CHP_ELEC_{TOTAL}	=	total CHP electricity generated at DES plant

For CHP plants in which a portion of the recovered heat is used to drive absorption chillers that provide cooling through a DES chilled-water loop, or a portion of the recovered heat is used for a third, separate district energy source (e.g., if the building connects to both a steam loop and a hot-water loop), calculate the electricity generation assigned to each building using the formula for heat recovery-driven chillers.

$$CHP\_ELEC_{BLDG} \text{ (heat recovery-driven chillers)} = (X_{HEAT} \times BLDG_{HEAT}) + (Y_{CHW} \times BLDG_{CHW}) + (Z_{SOURCE} \times BLDG_{SOURCE}) \times CHP\_ELEC_{TOTAL}$$

where

CHP_ELEC_{BLDG}	=	CHP electricity generation allocated to building
X_{HEAT}	=	fraction of CHP plant's total production of waste heat applied to the DES directly
$BLDG_{HEAT}$	=	fraction of total district heat provided to building
Y_{CHW}	=	fraction of CHP plant's total production of waste heat applied to producing chilled water in DES
$BLDG_{CHW}$	=	fraction of total district chilled water provided to building
Z_{SOURCE}	=	fraction of CHP plant's total production of waste heat applied to third district energy source
$BLDG_{SOURCE}$	=	fraction of third district energy source provided to building
CHP_ELEC_{TOTAL}	=	total CHP electricity generated at DES plant

When modeling CHP fuel input, allocate the CHP input fuel to the project building based on a proration and assignment of the total input fuel according to the results of the CHP electricity allocation described above for CHP electricity output. Use the energy cost and greenhouse gas emissions factors associated with the fuels input to the CHP. For the proposed case (all projects), calculate the CHP input fuel allocated to the building as follows:

$$\text{Proposed } BLDG_{FUEL} = \left(\frac{CHP\_ELEC_{BLDG}}{CHP\_ELEC_{TOTAL}} \right) \times CHP_{FUEL}$$

where

Proposed $BLDG_{FUEL}$	=	proposed case CHP input fuel allocated to building
CHP_ELEC_{BLDG}	=	CHP electricity generation allocated to building (from previous calculations)
CHP_ELEC_{TOTAL}	=	total CHP electricity generated at DES plant
CHP_{FUEL}	=	total CHP fuel input for electricity generation at DES plant

For the baseline (scenario B in CHP electricity output only): calculate the CHP input fuel allocated to the building as follows:

$$\text{Baseline BLDG}_{FUEL} = \left(\frac{\text{PROCESS\_ELEC}_{BLDG}}{\text{CHP\_ELEC}_{TOTAL}} \right) \times \text{CHP}_{FUEL}$$

with

$$\text{PROCESS\_ELEC}_{BLDG} = \text{CHP\_ELEC}_{BLDG} - \text{PROPOSED\_ELEC}_{BLDG}$$

where

$\text{Baseline BLDG}_{FUEL}$	=	baseline case CHP input fuel charged to building
$\text{PROCESS\_ELEC}_{BLDG}$	=	amount of allocated CHP electricity in excess of building's modeled annual electricity consumption (treated as process energy in model)
CHP_ELEC_{TOTAL}	=	total CHP electricity generated at DES plant
CHP_{FUEL}	=	total CHP fuel input for electricity generation at DES plant
CHP_ELEC_{BLDG}	=	CHP electricity generation allocated to building (from previous calculations)
$\text{PROPOSED\_ELEC}_{BLDG}$	=	modeled electricity consumption for building from proposed case

The model must include CHP generator efficiencies, based on either ongoing operations (existing CHP) or design specifications (new CHP).

Path 3 Large-scale District Energy System

Path 3 provides a streamlined method for documenting improved greenhouse gas (GHG) emissions performance associated with large scale district energy systems.

Complete the baseline and proposed energy modeling consistent with the guidance for Path 1A. ASHRAE 90.1–2016, Appendix G.

Follow the additional instructions in the *Large-Scale DES Calculator* (uploaded in the Credit Resources section of the credit library) to demonstrate GHG emissions improvement associated with the district energy plant.

Optional: Projects may generate two sets of baseline and proposed models to separately document the greenhouse gas emissions metric using Path 1A. ASHRAE 90.1-2016 Appendix G with the large-scale district energy calculator, and the cost metric using Path 1B. ASHRAE 90.1-2016 Appendix G with ASHRAE 90.1-2022 Addendum a.

Special Situations for DES Energy Models

Service water heating
If service water is heated in full or in part by DES-supplied heat: For projects applying Path 1A or Path 3, model the energy source as purchased energy.

Projects applying Path 1B shall model the DES supplied service water heating in the Proposed building per 90.1-2022 addendum a replacing all ASHRAE 90.1-2022 Addendum a references to Section 7 mandatory and prescriptive criteria for the *proposed building design* with ASHRAE 90.1-2016 Section 7.

The Baseline shall be modeled per 90.1-2016 Appendix G requirements.

For projects applying Path 2, model the baseline service water heating matching ASHRAE 90.1-2016 Appendix G modeling guidance for a stand-alone on-site service water heating system, and use the Path 2 guidance to model the average efficiency for the proposed design.

Heating converted to cooling
Sometimes the district or campus system heating energy supply is converted to chilled water using absorption chillers or other similar technologies to serve cooling loads. In this circumstance, the equipment that converts heating to cooling may reside within the DES itself, (i.e., DES provides cooling to the building) or within the connected buildings (i.e., DES provides heating to the building; building converts heating to cooling). When the equipment that converts DES-supplied heat into cooling is part of the LEED project's scope of work, the project must apply either Path 1B, Path 2, or Path 3.

Other DES systems

DES also often incorporate special features, such as thermal storage, ground or surface water cooling, and waste heat recovery. These features should be incorporated into the proposed virtual plant to the greatest extent practical using the general principles presented in this guidance.

Combined Heat and Power (CHP) or other Non-Renewable Electricity Generation Systems

For projects with combined heat and power or other non-renewable electricity generation systems, amend ASHRAE 90.1-2016 G2.4.2 Annual Energy Costs as follows:

Where the proposed design includes on-site electricity generation systems other than on-site renewable energy systems, adjust the baseline and proposed model using one of the following methods:

1. No credit for on-site electricity generation:
    - Model on-site electricity generation systems including all fuel inputs and associated site-recovered energy identically in the baseline design and the proposed design, OR
    - Model purchased electricity instead of the on-site electricity generation. Model any site-recovered energy from the on-site electricity generation system identically in the baseline and proposed design (either crediting it towards the thermal loads for both the baseline and proposed design or ignoring the site-recovered energy contribution in both the baseline and proposed design).
2. Credit for on-site electricity generation:
   Model the baseline design using purchased electricity for all regulated energy sources except HVAC heating and/or service water heating modeled in accordance with Appendix G criteria. Model the proposed design to include the proposed on-site generation system including site-recovered energy. For the cost metric, natural gas or fuel rates for both the baseline and proposed design must be modeled using the current published rates for natural gas associated with the baseline design fuel usage excluding monthly meter charges and shall not be discounted for high fuel usage associated with on-site generation equipment.

International Tips
Option 1. Energy Performance Compliance

Canada: Use the Provincial emissions factors (where available) reported in the National Inventory Report, submitted by Canada to the United Nations Framework Convention on Climate Change, to calculate GHG emissions by energy source; these emissions factors are readily found in the ENERGY

STAR Portfolio Manager Greenhouse Gas Emissions Technical Reference (https://portfoliomanager.energystar.gov/pdf/reference/Emissions.pdf).

Europe: The Europe ACP for ASHRAE 90.1 Mandatory Provisions Table provides further guidance for project teams in Europe wishing to use European standards in lieu of certain ASHRAE 90.1-2016 mandatory provisions in LEED v4.1. The guidance covers ASHRAE 90.1-2016 Mandatory Provision Sections 5.4, 6.4, 7.4, 8.4, 9.4 and 10. Column 1 of the table references the specific subsection used in ASHRAE 90.1-2016. Column 2 displays the requirement as written in ASHRAE 90.1-2016. Column 3 outlines the compliance pathway available for European projects. Column 4 includes, in some cases, further information about the proposal, differences between the proposal and the ASHRAE requirement, or a reference to further documentation.

Additionally, for projects using the Performance Option for compliance with EA prerequisite Minimum Energy Performance and EA credit Optimize Energy Performance, the documentation must also use the calculated U-factor for fenestration products including windows and skylights based on either the LBNL Windows 6 program, or a simulation software program that approximates the NFRC rating methodologies. Alternatively, a narrative shall be provided supporting the claim that the fenestration U-factor used in the model is similar to the values that would be achieved using the NFRC rating. The CE-marked fenestration does not account for thermal bridging and seasonal performance in the same way as the NFRC rating, and when accounted for in the energy model, has been observed to lead to savings that exceed those claimed for the same fenestration rated under the NFRC ratings.

Referenced Standards

- ASHRAE 90.1–2016
- ASHRAE 50% Advanced Energy Design Guides
- ASHRAE 209-2018
- ANSI/ASHRAE/IES Standard 90.1-2016 Performance Rating Method Reference Manual, PNNL 2017
- Developing Performance Cost Index Targets for ASHRAE Standard 90.1 Appendix G - Performance Rating Method

Required Documentation

Documentation	Energy Performance Option		
	90.1-2016 Prescriptive Compliance	90.1-2016 ECB Compliance (prerequisite only)	90.1 Appendix G Compliance
Minimum Energy Performance Calculator (90.1-2016) with Appendix G energy modeling inputs			X
Input-output reports from modeling software		X	X
Exceptional calculations (if applicable)			X
Energy consumption and demand for each building end use and fuel type			X
Description of energy utility rates for each energy source		X	X (if using cost metric)
Greenhouse gas emissions calculations, including emissions factors used			X (if using GHG metric)

ASHRAE 90.1-2016 Compliance Forms or COMcheck report confirming a system-by-system compliance with all mandatory and prescriptive provisions, and supporting construction drawings	X		
Documentation demonstrating compliance with ASHRAE 90.1-2016 Mandatory Measures and ECB		X	
Data center calculator (if applicable)			X
On-site renewable energy plans indicating location of renewable energy system, and relevant design details (e.g. PV module capacity, quantity, inverter capacity, tilt, orientation, etc. for a photovoltaic array), and confirming that the renewable energy is part of the project scope of work (or campus scope of work for a campus development)			X

Connection to Ongoing Building Performance

- LEED O+M EA credit Energy Performance: Designing building systems to achieve a minimum level of energy efficiency provides the foundation for effective energy management, reduced greenhouse gas emissions from building energy use, and reduced operating costs throughout the building life cycle.